

CSA1713 – Artificial Intelligence

Name:- B. Deepthi reddy

Reg No:- 192312288

Slot – C

9. CRYPT ARITHMETIC PROBLEM

Code:

```
print("B. Deepthi Reddy")
print(192312288)
print("CRYPT ARITHMETIC PROBLEM")

from itertools import permutations

for p in permutations(range(10), 8):
    S, E, N, D, M, O, R, Y = p

    if S != 0 and M != 0:
        send = 1000 * S + 100 * E + 10 * N + D
        more = 1000 * M + 100 * O + 10 * R + E
        money = 10000 * M + 1000 * O + 100 * N + 10 * E + Y

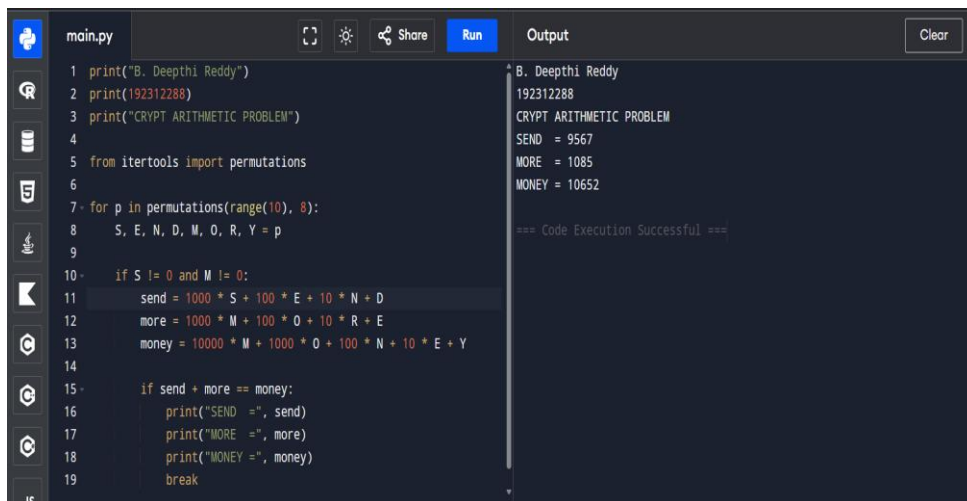
        if send + more == money:
            print("SEND =", send)
```

```
print("MORE =", more)

print("MONEY =", money)

break
```

Output:



The screenshot shows a code editor with a file named 'main.py'. The code is a Python script that prints the author's name and a problem number, then iterates through permutations of the digits 0-9 to solve a cryptarithm. The output window on the right shows the execution results.

```
main.py  [Run] [Share] [Clear]

1 print("B. Deepthi Reddy")
2 print(192312288)
3 print("CRYPT ARITHMETIC PROBLEM")
4
5 from itertools import permutations
6
7 for p in permutations(range(10), 8):
8     S, E, N, D, M, O, R, Y = p
9
10    if S != 0 and M != 0:
11        send = 1000 * S + 100 * E + 10 * N + D
12        more = 1000 * M + 100 * O + 10 * R + E
13        money = 10000 * M + 1000 * O + 100 * N + 10 * E + Y
14
15        if send + more == money:
16            print("SEND =", send)
17            print("MORE =", more)
18            print("MONEY =", money)
19            break
```

Output:

```
B. Deepthi Reddy
192312288
CRYPT ARITHMETIC PROBLEM
SEND  = 9567
MORE  = 1085
MONEY = 10652

=== Code Execution Successful ===
```